

Reg No.: \_\_\_\_\_

Name: \_\_\_\_\_

**APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY**  
Second Semester B.Tech Degree Examination July 2021 (2019 scheme)

**Course Code: EST102**  
**Course Name: PROGRAMMING IN C (Common to all programs)**  
**(FN Session)**

Max. Marks: 100

Duration: 3 Hours

**PART A**

*Answer all Questions. Each question carries 3 Marks*

Marks

- |    |                                                                                                                                                           |     |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1  | Differentiate between system software and application software.                                                                                           | (3) |
| 2  | Differentiate between compiler and interpreter.                                                                                                           | (3) |
| 3  | What is the importance of precedence and associativity? Write the table for operator precedence.                                                          | (3) |
| 4  | Differentiate between 'break' and 'continue' statements.                                                                                                  | (3) |
| 5  | Explain any 3 string handling functions using examples.                                                                                                   | (3) |
| 6  | Write a C program to find the occurrence of each element in an array.                                                                                     | (3) |
| 7  | Define formal parameters and actual parameters. Illustrate with an example.                                                                               | (3) |
| 8  | With examples show how:<br>(i) an array is passed as argument of a function.<br>(ii) individual elements of an array is passed as argument of a function. | (3) |
| 9  | Write any three file handling functions in C.                                                                                                             | (3) |
| 10 | Differentiate between address operator(&) and indirection(*) operator.                                                                                    | (3) |

**PART B**

*Answer any one Question from each module. Each question carries 14 Marks*

- |    |                                                          |     |
|----|----------------------------------------------------------|-----|
| 11 | a) Explain different types of memory used in a computer. | (7) |
|    | b) Write an algorithm to find sum of digits of a number. | (7) |

**OR**

- |    |                                                                                                                                                                              |      |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 12 | Explain bubble sort with an example. Draw a flowchart and write pseudo code to perform bubble sort on an array of numbers.                                                   | (14) |
| 13 | a) Explain different data types supported by C language with their memory requirements.                                                                                      | (7)  |
|    | b) Write a C program to check if a number is present in a given list of numbers. If present, give location of the number otherwise insert the number in the list at the end. | (7)  |

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**OR**

- 14 a) Write a C program to find the sum of first and last digit of a number. (7)  
b) What is type casting? Name the inbuilt typecasting functions available in C language. What is the difference between type casting and type conversion? (7)
- 15 a) Write a C program to perform linear search on an array of numbers. (7)  
b) Write a C program to reverse a string without using string handling functions. (7)

**OR**

- 16 a) Write a C program to find the transpose of a matrix. (7)  
b) Write a C program to print number of vowels and consonants in a string. (7)
- 17 a) What is the purpose of function declaration and function definition and function call? With examples illustrate their syntax. (7)  
b) Write a C program to : (7)  
(i) Create a structure containing the fields: Name, Price, Quantity, Total Amount.  
(ii) Use separate functions to read and print the data

**OR**

- 18 a) What are different storage classes in C? Give examples for each. (7)  
b) Write a C program to find sum and average of an array of integers using user defined functions. (7)
- 19 a) Explain the different modes of operations performed on a file in C language. (7)  
b) Write a program in C to copy the contents of one file into another. (7)

**OR**

- 20 a) Explain how pointers can be passed to functions in C. (7)  
b) Explain any 5 file handling functions in C? (7)

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